

120 Days of Quant Finance

Week 3 – Linear Algebra (Finance)

Day 6: Risk Attribution Using Linear Algebra

Arnav Garg

January 2026

Introduction

Risk attribution seeks to explain how individual assets or factors contribute to overall portfolio risk. Using linear algebra, portfolio risk can be decomposed systematically into additive components, enabling precise risk management and portfolio control.

Portfolio Risk

Let $\mathbf{w} \in \mathbb{R}^n$ denote portfolio weights and $\Sigma \in \mathbb{R}^{n \times n}$ the covariance matrix of asset returns. Portfolio variance is given by:

$$\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}.$$

Marginal Contribution to Risk

The marginal contribution to portfolio volatility is defined as:

$$\text{MCR}_i = \frac{\partial \sigma_p}{\partial w_i}.$$

In vector form:

$$\text{MCR} = \frac{\Sigma \mathbf{w}}{\sigma_p}.$$

Total Risk Contribution

The total risk contribution of asset i is:

$$\text{TRC}_i = w_i \cdot \text{MCR}_i.$$

In vector notation:

$$\mathbf{TRC} = \mathbf{w} \odot \frac{\mathbf{\Sigma} \mathbf{w}}{\sigma_p},$$

where $\odot$ denotes element-wise multiplication.

The total portfolio risk satisfies:

$$\sum_{i=1}^n \text{TRC}_i = \sigma_p.$$

Factor Model Risk Decomposition

Assuming a linear factor model:

$$\mathbf{\Sigma} = \mathbf{B} \mathbf{\Sigma}_f \mathbf{B}^\top + \mathbf{\Sigma}_\varepsilon,$$

portfolio variance decomposes as:

$$\sigma_p^2 = \mathbf{w}^\top \mathbf{B} \mathbf{\Sigma}_f \mathbf{B}^\top \mathbf{w} + \mathbf{w}^\top \mathbf{\Sigma}_\varepsilon \mathbf{w}.$$

Factor Risk Attribution

Portfolio factor exposure is defined as:

$$\boldsymbol{\beta}_p = \mathbf{B}^\top \mathbf{w}.$$

Factor risk contribution is given by:

$$\sigma_{\text{factor}}^2 = \boldsymbol{\beta}_p^\top \mathbf{\Sigma}_f \boldsymbol{\beta}_p.$$

PCA-Based Risk Attribution

Using eigen-decomposition:

$$\mathbf{\Sigma} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top,$$

portfolio variance can be written as:

$$\sigma_p^2 = \sum_{k=1}^n \lambda_k (\mathbf{q}_k^\top \mathbf{w})^2.$$

Each term represents the contribution of an independent risk direction.

Conclusion

Risk attribution reframes portfolio management in terms of contributions rather than positions. Linear algebra provides a rigorous framework for decomposing portfolio risk into asset-level, factor-level, and principal-component-level components, forming the backbone of professional risk management systems.